@PTKT

SaigonBite

Introduction to Software Engineering (SE) – Sem01_2627

Advised by ThS. Phạm Thái Kỳ Trung

By Students:

1. 524H0195 : Ngô Lâm Tiến
2. 524H0191 : Nguyễn Công Thành
3. 524H0179 : Nguyễn Trọng Phú

Team	Chosen tier	Document version	Date
TeamG4	Tier 4 proposed Ceiling 10.0/10	v1.0	25/10/2026

Abstract

SaigonBite addresses inconsistent prices, fragmented ordering and uneven hub workloads in Ho Chi Minh City's cloud kitchens. TeamG4 proposes an MVC application with explicit use case ownership, deterministic pricing and a controlled order lifecycle. This Part 1 baseline specifies 47 functional requirements, six quality attributes and five working assumptions, with planned traceability. Tier 4 is proposed through a Code-First foundation, live payment sandbox, interactive Admin dashboard and automated tests; implementation and execution evidence are later-phase deliverables.

Intellectual Property

Copyright 2026 TeamG4.

The following documentation, the content therein and/or the presentation of its information is proprietary to and embodies the confidential processes, designs, technologies and otherwise of TeamG4. All copyright, trademarks, trade names, patents, industrial designs, and other intellectual property rights contained herein are, unless otherwise specified, the exclusive property of TeamG4.

The ideas, concepts and/or their application, embodied within this documentation remain and constitute items of intellectual property which nevertheless belong to TeamG4.

The information (including, but by no means limited to, data, drawings, specification, documentation, software listings, source and/or object code) shall not be disclosed, manipulated, disseminated or otherwise in any manner inconsistent with the nature and/or conditions under which this documentation has been issued.

The information contained herein is believed to be accurate and reliable. TeamG4 accepts no responsibility for its use in any way whatsoever. TeamG4 shall not be liable for any expenses, damages and/or related costs which may result from the use of the information contained herein.

The information contained herein is subject to change without notice.

All Rights Reserved. Copyright herein is expressly protected at common law, statute and under various International and Multi-National Treatises (including, but by no means limited to, the Berne Convention for the Protection of Literary and Artistic Works).

Table of Contents

Intellectual Property....2
Table of Contents....3
Feedback Log....4
Part 1: Proposal & Requirements Engineering (15 marks)....5
1.1 Project Proposal....5
1.2 Problem Statement....7
1.3 Functional Requirements....7
1.4 Quality Attributes (Non-Functional Requirements)....11
1.5 Ambiguity Log....12
References....13
Appendix....14
Appendix A: Contribution Statement....14
Appendix B: Generative AI Usage Log....14
Appendix C: Requirements Traceability Matrix....15

Feedback Log

Source	Feedback received	What we changed and where	Done
Presentation 1	Not yet received; presentation pending.	No instructor feedback is claimed in this submission.	Pending
Part 1 marking	Not yet received; marking pending.	Record the actual comments and resulting changes after marking.	Pending

Part 1: Proposal & Requirements Engineering (15 marks)

1.1 Project Proposal

Methodology and team roles

We propose Scrum in a seven-week plan, using one-week planning cycles, a short daily coordination meeting, a weekly integration review and a retrospective after each milestone. Small cycles expose integration risks early while preserving the Part 1, Part 2 and Part 3 submission milestones. The Week 5 change request enters the backlog with an explicit impact assessment.

No.	Name and student ID	Role	Main responsibilities
1	Ngô Lâm Tiến 524H0195	Team leader; pricing and payment integration	C - Manage Vouchers; Pricing Engine (BR1-BR12); Payment Strategy Module
2	Nguyễn Công Thành 524H0191	Data model, catalogue and Revenue report	E - Manage User Accounts; R1 - Revenue Report
3	Nguyễn Trọng Phú 524H0179	Hub capacity, dispatch and Hub Utilization report	F - Manage Deliveries; Change Request Impact Note; R2 - Hub Utilization Report

Use case ownership

No.	Main use case (CRUD)	Owner (student ID)	Candidate list item (A to F) or approved own proposal
UC1	Place and Manage Orders	524H0195	D
UC2	Manage Menu & Brands	524H0191	A
UC3	Manage Hubs & Capacity	524H0179	B

UC1 (D) belongs to Ngô Lâm Tiến: create an order; read/track owned orders; edit items only while Placed; cancel while Placed or Confirmed, retaining history. UC2 (A) belongs to Nguyễn Công Thành: create, read, update and soft-delete brands/menu items. UC3 (B) belongs to Nguyễn Trọng Phú: create, read, update and soft-delete hubs/capacity while retaining order history. Every member must speak and claim this main use case in Presentation 1.

Schedule and submission milestones

Period	Milestone	Planned output
Weeks 1–2	26 Oct 2026, 07:00 UTC+7	Part 1 report, tracking workbook and 5–7 minute Solution Pitch.
Weeks 3–4	9 Nov 2026, 07:00 UTC+7	Part 2 modelling, architecture and Figma prototype.
Weeks 5–7	30 Nov 2026, 07:00 UTC+7	Running skeleton, change-request analysis, implementation, QA and Part 3 demo.
Final consolidation	7 Dec 2026	Survival Guide date; follow the final portal instruction when published.

Technology tier and acceptance plan

Tier	Ceiling /10	Required stack and architecture
Tier 1	7.0	C# WinForms + local SQL database / Excel. 3-Tier architecture.
Tier 1B	7.5	Core PHP (or CodeIgniter) + MySQL + Bootstrap. MVC architecture.
Tier 2	8.5	ASP.NET Web Forms/MVC (Database First) + SQL Server + Bootstrap. MVC.
Tier 3	9.5	ASP.NET MVC (Code First) + Entity Framework + Bootstrap. MVC.
Tier 4	10.0	Tier 3 plus live sandbox API, interactive Admin dashboard and automated tests.

The proposed target is Tier 4, ceiling 10.0/10 under the instructor table supplied on 24 October 2026. The cap limits the rubric result; it is not an awarded mark. We propose ASP.NET Core MVC, EF Core Code-First, SQL Server and Bootstrap 5. The source names ASP.NET MVC/Entity Framework, so acceptance of the Core variant remains pending. SQL Server is the submission baseline; SQLite is only a labelled local fallback.

Tier 3 foundation

ASP.NET Core MVC, EF Core Code-First migrations, SQL Server and Bootstrap 5. Retain the MVC layers; confirm the Core variant with the instructor because the table names ASP.NET MVC.

Owner: 524H0191. Evidence: Part 2 architecture/ERD and migration design; Part 3 reproducible SQL Server setup, migration history and complete CRUD demonstration. SQLite is a local fallback, not the submitted database baseline. Trace: UC A/B/D; NFR3/4. Status: Planned.

Live sandbox API

Use the VNPay sandbox for actor-eligible orders. Validate callbacks and order/amount; prevent duplicate settlement. Keep mock payments explicitly labelled.

Owner: 524H0195. Evidence: Part 3 redacted provider transaction IDs and callback logs for success, cancellation/failure, invalid signature and repeated callback; automated TC-P-05–09. Credentials are excluded. Mock-only evidence does not meet this criterion. Trace: FR-P1/P3/P4; FR-S5. Status: Planned.

Interactive Admin dashboard

Provide R1 Revenue and R2 Hub Utilization with charts and tables. Revenue date-range and hub-selection controls refresh their respective views from authorised server data.

Owner: 524H0191 / 524H0179. Evidence: Part 2 dashboard prototype; Part 3 recorded filter interaction, chart/table reconciliation against seeded data, and denied non-Admin access. Automated TC-S-03a–c. Trace: FR-R1/R2; FR-S2/S3. Status: Planned.

Automated tests

Run unit tests for pricing/state rules and integration/E2E tests for orders, sandbox callbacks and the Admin dashboard. Cover at least 20 EP/BVA cases and five forbidden transitions.

Owner: 524H0195 / 524H0191 / 524H0179. Evidence: Part 3 versioned commands and fixtures, commit identifier and execution reports with counts and PASS/FAIL outcomes. Planned test IDs and screenshots alone do not establish execution. Trace: RTM; TC-P-05–09; TC-S-03a–c. Status: Planned.

.NET 8 is the existing baseline and reaches end of support on 10 November 2026, before later milestones. Record this lifecycle risk and agree any runtime upgrade with the instructor. Arrange sandbox credentials early. A simulated payment is an explicitly labelled fallback and does not meet the live-sandbox criterion.

1.2 Problem Statement

Customers need one reliable way to browse food across brands and hubs, understand the complete price before paying, and track delivery. Kitchen staff must see orders belonging to their hub, control preparation and avoid accepting work when capacity is saturated. Administrators need consistent Revenue and Hub Utilization reports; shippers need authorised delivery transitions and COD confirmation. Local Customers and Foreign Tourists require different identity and payment options, while sharing the same pricing and order lifecycle. SaigonBite therefore needs traceable requirements that reconcile transparent charging, hub capacity, actor permissions and retained order history.

This Part 1 submission defines the proposal and requirements baseline. Detailed UML, Figma, source implementation, sandbox transactions and application test results belong to later deliverables. Planned components, tests and ownership are not completed execution evidence.

1.3 Functional Requirements

Amounts are VND; time windows use Vietnam local time (UTC+7), start inclusive and end exclusive. UC1=D, UC2=A and UC3=B; secondary scopes C/E/F remain assigned to these owners. R1 and R2 are mandatory. Each stable FR ID below maps to its source and use case; the seeded RTM adds owner/component/test mappings.

ID	Requirement ("The system shall…")	Use case	Business rules	Report
FR-A1	The system shall let an administrator create a brand (name, description, operating hubs) and create its menu items (name, category, positive VND price, brand and availability).	A	BR12	—
FR-A2	The system shall let any visitor view brands and menu items filtered by hub and by category.	A	BR12	—
FR-A3	The system shall let an administrator edit brand details and a menu item’s name, category, price and brand.	A	BR12	—
FR-A4	The system shall soft-delete brands and menu items, so that historical orders and reports keep their names and prices.	A	BR12	—
FR-A5	The system shall let an administrator set item availability; unavailable or inactive items shall not be orderable.	A	BR12	—
FR-A6	The system shall reject any order line whose item belongs to a brand the chosen hub does not operate.	A	BR12	—
FR-B1	The system shall let an administrator create a hub with a district, an address and a whole-number kitchen capacity greater than zero.	B	BR8	—

ID	Requirement ("The system shall…")	Use case	Business rules	Report
FR-B2	The system shall display hub load as 100 times the count of Confirmed or Preparing orders divided by positive integer kitchen capacity; the new order is not counted.	B	BR8	—
FR-B3	The system shall let an administrator edit a hub's district, address and positive integer capacity, and recompute displayed load after a capacity change.	B	BR8	—
FR-B4	The system shall soft-delete a hub, removing it from ordering while preserving its historical orders and report rows.	B	BR8	—
FR-B5	The system shall refuse a new order when the chosen hub's load is 90 % or above and shall ask the customer to choose another hub.	B	BR8	—
FR-C1	The system shall let an administrator create a voucher with a code, a type (PERCENT or FIXED), a value, a maximum discount, a minimum subtotal, validity dates and an active flag.	C	BR9	—
FR-C2	The system shall let an administrator view, edit and deactivate vouchers.	C	BR9	—
FR-C3	The system shall accept at most one active voucher within its validity dates and check its minimum-subtotal condition against S before discounts.	C	BR9	—
FR-D1	The system shall let a customer build an order from the menus of one hub, with 1 to 30 lines and a quantity of 1 to 20 per line.	D	BR12	—
FR-D2	The system shall reject an order whose item subtotal is below 50,000.	D	BR1	—
FR-D3	The system shall reject an order whose delivery distance exceeds 8.0 km.	D	BR2	—
FR-D4	The system shall charge D = 15,000 for $d \le 2$ km; 15,000 + 5,000 × ceil(d − 2) for $2 < d \le 5$; and 30,000 + 7,000 × ceil(d − 5) for $5 < d \le 8$.	D	BR3	—
FR-D5	The system shall charge R = 20% of non-waived D when placement time in Vietnam (UTC+7) is in [11:00, 13:00) or [17:30, 19:30); otherwise R = 0.	D	BR4	—
FR-D6	The system shall add a flat 10,000 rain surcharge while the administrator's heavy-rain flag is on, using the flag value snapshotted on the order.	D	BR5	—
FR-D7	The system shall waive the delivery fee and the rush surcharge when the item subtotal is 300,000 or more, while the rain surcharge still applies.	D	BR6	—

ID	Requirement ("The system shall…")	Use case	Business rules	Report
FR-D8	The system shall apply a Quiet Kitchen discount of 10 % of the item subtotal when the hub load at placement is below 30 %.	D	BR7	—
FR-D9	The system shall calculate PERCENT voucher V on S − Q, limited by the voucher cap and S − Q; FIXED V shall not exceed S − Q.	D	BR9	—
FR-D10	The system shall calculate S, Q, V, F = S − Q − V, D, R, R2 and raw total = F + D + R + R2 in BR10 order, retain fractional values without intermediate rounding or truncation, then round once to the nearest 1,000 VND, half up.	D	BR10, BR11	—
FR-D11	The system shall show an itemised price breakdown — subtotal, Quiet Kitchen discount, voucher, food total, delivery fee, rush surcharge, rain surcharge, rounded total — before payment.	D	BR10	—
FR-D12	The system shall allow items to be edited only while the order is Placed, and any recalculation shall reuse the stored placement snapshot.	D	BR17	—
FR-D13	The system shall allow Customer cancellation in Placed or Confirmed, Kitchen Staff rejection in Placed with a reason, and Admin cancellation in Confirmed; online-paid cancellations shall be refunded in full (simulated).	D	BR16	—
FR-D14	The system shall store a price snapshot of timestamp, hub load and rain flag at placement so that the price stays reproducible.	D	BR17	—
FR-E1	The system shall register a Local Customer with a Vietnamese mobile number of exactly ten digits starting 03, 05, 07, 08 or 09, and a simulated OTP.	E	BR18	—
FR-E2	The system shall register a Foreign Tourist with a passport number of 6 to 9 letters or digits, a nationality, a valid email and a simulated email code.	E	BR18	—
FR-E3	The system shall enforce a unique identifier and a password of at least eight characters stored as a hash.	E	BR18	—
FR-E4	The system shall store both customer types in one schema using a discriminator, and let an administrator view, edit and deactivate accounts.	E	BR18	—
FR-F1	The system shall let an administrator assign an available shipper to a Ready order and record the shipper and assignment time (assumption A1).	F	BR15	—

ID	Requirement ("The system shall…")	Use case	Business rules	Report
FR-F2	The system shall enforce all eight transitions and actor/condition constraints in Specification section 2C in the business layer and reject every other transition; Delivered, Cancelled and Failed are terminal.	F	BR15	—
FR-F3	The system shall require a shipper to confirm cash collected before a COD order may become Delivered.	F	BR14	—
FR-F4	The system shall let the assigned shipper move Out for Delivery to terminal Failed when the customer is unreachable, recording a reason; financial settlement follows assumption A2.	F	BR15, BR16	—
FR-P1	The system shall offer only the payment methods that belong to the customer's actor group: MoMo, VNPay and cash for Local Customers; Apple Pay, PayPal and cash for Foreign Tourists.	D	BR13	—
FR-P2	The system shall refuse Cash on Delivery when the rounded total exceeds 1,000,000.	D	BR14	—
FR-P3	The system shall record each payment attempt with order, selected method, timestamp and success/failure outcome; only an actor-permitted Strategy may be selected.	D	BR13	—
FR-P4	The system shall integrate the live VNPay sandbox API for eligible Local Customer orders, authenticate callbacks and validate the order and amount before recording success, and process repeated callbacks idempotently.	D	Tier 4 instructor table; BR13	—
FR-R1	The system shall produce a Revenue report for a chosen date range showing total revenue by brand category and by payment method, excluding Cancelled and Failed orders.	A	R1	R1
FR-R2	The system shall produce a Hub Utilization report showing, per hub, orders handled, average preparation time from Confirmed to Ready, current load percentage and the number of Quiet Kitchen discounts granted.	B	R2	R2
FR-S1	The system shall let any user switch the interface between English and Vietnamese, and the choice shall persist for the session.	E	NFR3	—
FR-S2	The system shall restrict each role to its own functions on the server, not only in the interface.	E	NFR4	—

ID	Requirement ("The system shall…")	Use case	Business rules	Report
FR-S3	The system shall provide an interactive administrator dashboard for R1 and R2 with revenue date-range and hub-selection controls that refresh the respective charts and tables from the same authorised report data.	A/B	Tier 4 instructor table; R1, R2	R1, R2
FR-S4	The system shall let an administrator toggle the heavy-rain flag that the rain surcharge depends on.	D	BR5	—
FR-S5	The system shall append an audit record for each status change (actor, timestamp, old/new state) and payment attempt (actor, timestamp, method, outcome), without an application-level edit/delete operation.	D	BR15, NFR5	—

Order lifecycle and precision acceptance baseline

Allowed transitions are Placed→Confirmed (Kitchen Staff at the hub), Placed→Cancelled (Customer or hub staff with rejection reason), Confirmed→Preparing (hub staff), Confirmed→Cancelled (Customer/Admin), Preparing→Ready (hub staff), Ready→Out for Delivery (assigned Shipper), Out for Delivery→Delivered (assigned Shipper; COD collected first) and Out for Delivery→Failed (assigned Shipper; unreachable customer). Delivered, Cancelled and Failed are terminal; every other transition is forbidden. Source: specification section 2C and BR14–BR16.

Preserve fractional discounts until the final half-up rounding (BR10/BR11). Acceptance example: subtotal 61,111; load 20%; valid uncapped 10% voucher; 1 km; no rush/rain. Quiet Kitchen discount=6,111.1; voucher=5,499.99; raw=64,499.91; final=64,000. Intermediate truncation would incorrectly yield 65,000. These are independent expected values, not application test results.

1.4 Quality Attributes (Non-Functional Requirements)

NFR1–NFR4 refine the mandatory performance, maintainability, usability and security baselines; NFR5/NFR6 are team-defined. All verification is planned. Each target below retains its source and owner in the RTM.

ID	Attribute	Measurable target	How you will verify it
NFR1	Performance	Price calculation for a valid order of 30 lines completes in under 1.5 seconds server-side.	Time the engine with 30 valid lines on the documented test machine; record environment, elapsed times and pass/fail in Part 3.
NFR2	Maintainability	Use MVC separation. Adding a payment method requires one new Strategy class and one registration, with zero edits to existing payment classes.	Add a sample method in a disposable branch; inspect the diff and rerun existing payment tests. This criterion applies to a payment-extension change request.

ID	Attribute	Measurable target	How you will verify it
NFR3	Usability	All required screens support English/Vietnamese and operate without horizontal page scrolling at 390, 768 and 1440 px; culture switching preserves the cart and session.	Check each screen and language at the three widths, verify resource keys and repeat switching with a non-empty cart.
NFR4	Security	Passwords are hashed, never stored as plaintext; every role is refused every server action outside its authorised role and resource scope.	Inspect stored password representation; run an access matrix for both customer types, Kitchen Staff, Shipper and Admin, including other users' orders and hubs.
NFR5	Auditability	100% of accepted status changes and payment attempts append a record with actor and timestamp; status changes include old/new state, payment attempts include method/outcome. No application update/delete endpoint is provided.	Reconcile accepted transitions and payment attempts against audit rows and verify ordinary roles cannot change recorded history.
NFR6	Reliability	Zero forbidden transitions are accepted; repeated pricing with identical lines and the stored placement snapshot returns identical components and rounded total.	Enumerate the state/actor matrix, include at least five forbidden transitions, and repeat boundary pricing inputs against a fixed snapshot.

Sources and ownership: NFR1: Case Study section 4B; BR12; owner 524H0195; NFR2: Case Study section 4B; BR13; owner 524H0195; NFR3: Case Study section 4B; owner 524H0191; NFR4: Case Study section 4B; BR18; owner 524H0191; NFR5: Team-defined; BR15; R2; owner 524H0195 / 524H0179; NFR6: Team-defined; BR10, BR11, BR15, BR17; owner 524H0195 / 524H0179.

1.5 Ambiguity Log

These are working assumptions, not instructor approvals. Fixed thresholds, calculation order and authorised state transitions are not open ambiguities. A changed answer requires updating the affected requirements, design and tests.

ID	Ambiguity or gap	Assumption we made	Instructor answer / date
A1	Shipper assignment: Section 2C leaves the assignment mechanism open. Impact: FR-F1; UC F; assignment screen and service.	An Admin manually selects an available shipper for a Ready order. The system stores assignment time and shipper. Automatic dispatch is outside the initial scope.	Not asked; working assumption pending instructor review.
A2	Settlement after Failed delivery: Section 2C leaves the financial result of Failed delivery open. Impact: FR-F4; UC F/D; payment/refund policy and R1.	Refund online-paid orders in full in the simulated ledger; collect no COD cash. Preserve Failed as its own terminal state and exclude it from R1 revenue.	Not asked; working assumption pending instructor review.

ID	Ambiguity or gap	Assumption we made	Instructor answer / date
A3	Tourist delivery address: Hotel and room are allowed, but a field structure is not prescribed. Impact: FR-E2 and FR-D1; address validation and checkout.	Store street/address text plus optional hotel name and room. Require a usable delivery address for either customer type; tourists are not required to supply a Vietnamese phone.	Not asked; working assumption pending instructor review.
A4	Invalid voucher user flow: BR9 specifies eligibility but not whether a refused voucher should block the whole order. Impact: FR-C3 and FR-D9/D11; checkout confirmation.	Show the refusal and the corrected breakdown with V=0; require the customer to review and confirm that total before placing the order. Never silently submit a higher total.	Not asked; working assumption pending instructor review.
A5	Catalogue price change before a Placed edit: BR17 fixes timestamp/load/rain but does not explicitly say which item prices apply when the catalogue changes before an edit. Impact: FR-D12/D14 and FR-A4; order-line price storage and edit tests.	Existing lines retain their recorded unit prices; newly added items use the current catalogue price after availability/membership validation. Recalculation still uses the original timestamp, load and rain snapshot.	Not asked; working assumption pending instructor review.

References

1. Ton Duc Thang University, Faculty of Information Technology (2026). SE Final Project Specification: SaigonBite, version 1.0, Sem01_2627. BR1–BR18, Part 1 checklist and rubric.

2. Ton Duc Thang University (2026). SaigonBite Case Study: Cloud-Kitchen Food Delivery Hub. Context, FR1–FR9 and quality baselines.

3. Ton Duc Thang University (2026). SaigonBite Student Survival Guide. Milestones, checklists and presentation guidance.

4. Course instructor (2026). SE_Sem01_2627_Report_Template.docx. Required report structure, Part 1 sections 1.1–1.5 and appendices. Original retained unchanged with the course source files.

5. Course instructor. Technology tiers (from the specification; instructor may edit the blue caps). Image supplied by the student on 24 October 2026; publication date not shown. Archived at docs/references/instructor-technology-tiers.png.

6. Schwaber, K. and Sutherland, J. (2020). The Scrum Guide. https://scrumguides.org/scrum-guide.html. Basis for the proposed iterative process.

7. Microsoft (2026). .NET 8 and .NET 9 will reach End of Support on November 10, 2026. https://devblogs.microsoft.com/dotnet/dotnet-8-9-end-of-support/. Runtime lifecycle risk.

Appendix

Appendix A: Contribution Statement

The following responsibilities are planned. The course requires signatures from all members at the start of Part 3. Members must supply their own signatures, dates and actual contributions; no signature is generated in this Part 1 submission.

Student (ID)	Use case(s) owned	Classes and components	Signature and date
Ngô Lâm Tiến 524H0195	D — Place and Manage Orders	C - Manage Vouchers; Pricing Engine (BR1-BR12); Payment Strategy Module	Signature: _____ Date: _____
Nguyễn Công Thành 524H0191	A — Manage Menu & Brands	E - Manage User Accounts; R1 - Revenue Report	Signature: _____ Date: _____
Nguyễn Trọng Phú 524H0179	B — Manage Hubs & Capacity	F - Manage Deliveries; Change Request Impact Note; R2 - Hub Utilization Report	Signature: _____ Date: _____

Appendix B: Generative AI Usage Log

AI assistance is disclosed below. Original full historical DSH prompts/outputs have not been located; recover the original exports for the instructor's 48-hour request. The available Codex record at docs/appendix/prompts/CODEX_2026-10-24.md is labelled excerpts and a work summary, not a full transcript.

Tool and date	Purpose	Prompt (summary)	What you kept or changed	How you verified it
DeepSeek Harness (DSH), 23 Oct 2026	Earlier draft and project scaffolding	Earlier project log describes extraction of BRs, FR drafting, architecture and rendering. This is a summary from that log, not a recovered raw prompt.	Retained the useful document structure and stable FR IDs; corrected conflicting thresholds, ownership and unsupported claims during this revision.	Compared current requirements with the three course PDFs; earlier raw prompts/outputs are not available in this workspace.
OpenAI Codex, 24 Oct 2026	Strict Part 1 audit	User requested a lecturer-style audit of existing Part 1 against the screenshot, course requirements, templates and TDTU skills.	Recorded faults and repair priorities in PART1_AUDIT_2026-10-24.md; did not use the provisional score as an official course mark.	Read the course PDFs, extracted workbook cells, rendered report/slides, checked links and calculated the rounding counterexample independently.
OpenAI Codex, 24 Oct 2026	Correct and synchronise Part 1 deliverables	User: "ok giờ bạn cứ sửa sao cho chuẩn nhất đi để hoàn thiện part 1".	One shared FR/NFR/ambiguity register, corrected report/workbook/pitch and explicit planned verification. No invented signatures or instructor answers.	Build and source-consistency checks; rendered output checks are recorded in the Part 1 verification note. Live pitch, sandbox and application tests are outside this document revision.

Appendix C: Requirements Traceability Matrix

This is the Part 1 seed for the RTM completed in template section 3.5 at Part 3. Every FR/NFR has a planned design, component, test and owner. All verdicts are Planned; no application run is certified.

ID	Use case	Design element	Code component	Test IDs	Verdict
FR-A1	A Owner 524H0191	Brand entity, menu admin	BrandsController	TC-A-01	Planned
FR-A2	A Owner 524H0191	Catalogue view	BrandsController	TC-A-02	Planned
FR-A3	A Owner 524H0191	Menu edit form	BrandsController	TC-A-03	Planned
FR-A4	A Owner 524H0191	Soft-delete filter	DbContext filters	TC-A-04	Planned
FR-A5	A Owner 524H0191	Availability flag	MenuItemService	TC-A-05	Planned
FR-A6	A Owner 524H0191	Hub-brand relation	OrderService	TC-A-06	Planned
FR-B1	B Owner 524H0179	Hub entity	HubsController	TC-B-01	Planned
FR-B2	B Owner 524H0179	Derived load	HubLoadCalculator	TC-B-02	Planned
FR-B3	B Owner 524H0179	Capacity edit	HubsController	TC-B-03	Planned
FR-B4	B Owner 524H0179	Soft-delete filter	DbContext filters	TC-B-04	Planned
FR-B5	B Owner 524H0179	Admission rule	OrderAdmissionService	TC-B-05	Planned
FR-C1	C Owner 524H0195	Voucher entity	VouchersController	TC-C-01	Planned
FR-C2	C Owner 524H0195	Voucher admin	VouchersController	TC-C-02	Planned
FR-C3	C Owner 524H0195	Voucher validation	VoucherService	TC-C-03	Planned
FR-D1	D Owner 524H0195	Cart, line limits	OrdersController	TC-D-01	Planned
FR-D2	D Owner 524H0195	Minimum order	PricingEngine	TC-D-02, 03	Planned
FR-D3	D Owner 524H0195	Coverage check	PricingEngine	TC-D-04, 05	Planned
FR-D4	D Owner 524H0195	Distance tiers	PricingEngine	TC-D-06…08	Planned

ID	Use case	Design element	Code component	Test IDs	Verdict
FR-D5	D Owner 524H0195	Rush windows	PricingEngine	TC-D-09…11	Planned
FR-D6	D Owner 524H0195	Rain surcharge	PricingEngine	TC-D-12, 13	Planned
FR-D7	D Owner 524H0195	Free-delivery waiver	PricingEngine	TC-D-14, 15	Planned
FR-D8	D Owner 524H0195	Quiet Kitchen rule	PricingEngine	TC-D-16, 17	Planned
FR-D9	D Owner 524H0195	Voucher application	PricingEngine	TC-D-18…20	Planned
FR-D10	D Owner 524H0195	Nine-step order	PricingEngine	TC-D-21, 22	Planned
FR-D11	D Owner 524H0195	Breakdown view	OrdersController	TC-D-23	Planned
FR-D12	D Owner 524H0195	Edit while Placed	OrderService	TC-D-24	Planned
FR-D13	D Owner 524H0195	Cancellation rules	OrderStateMachine	TC-D-25, 26	Planned
FR-D14	D Owner 524H0195	Price snapshot	Order entity	TC-D-27	Planned
FR-E1	E Owner 524H0191	Local validation	AccountController	TC-E-01, 02	Planned
FR-E2	E Owner 524H0191	Tourist validation	AccountController	TC-E-03, 04	Planned
FR-E3	E Owner 524H0191	Hash + uniqueness	AccountService	TC-E-05	Planned
FR-E4	E Owner 524H0191	Discriminator schema	DbContext	TC-E-06	Planned
FR-F1	F Owner 524H0179	Shipper assignment	DeliveriesController	TC-F-01	Planned
FR-F2	F Owner 524H0179	State machine	OrderStateMachine	TC-F-02, 03	Planned
FR-F3	F Owner 524H0179	Cash confirmation	DeliveryService	TC-F-04	Planned
FR-F4	F Owner 524H0179	Failure path	DeliveryService	TC-F-05	Planned
FR-P1	D Owner 524H0195	Actor-method map	PaymentStrategyResolver	TC-P-01, 02	Planned
FR-P2	D Owner 524H0195	COD ceiling	CodStrategy	TC-P-03	Planned

ID	Use case	Design element	Code component	Test IDs	Verdict
FR-P3	D Owner 524H0195	Payment record	PaymentService	TC-P-04	Planned
FR-P4	D Owner 524H0195	Sandbox adapter and callback validation	VnPayStrategy / PaymentService	TC-P-05–09	Planned
FR-R1	A Owner 524H0191	Revenue aggregation	ReportService	TC-RT-01, 02	Planned
FR-R2	B Owner 524H0179	Utilisation aggregation	ReportService	TC-RT-03, 04	Planned
FR-S1	E Owner 524H0191	Resource files	Localisation	TC-S-01	Planned
FR-S2	E Owner 524H0191	Authorisation policy	Policies	TC-S-02	Planned
FR-S3	A/B Owner 524H0191 / 524H0179	Dashboard	AdminController	TC-S-03a–c	Planned
FR-S4	D Owner 524H0195	Rain flag	AdminController	TC-S-04	Planned
FR-S5	D Owner 524H0195	Audit writer	OrderStatusHistory	TC-S-05	Planned
NFR1	D Owner 524H0195	Pure PricingEngine	PricingEngine	TC-NFR-1	Planned
NFR2	D Owner 524H0195	IPaymentStrategy / DI	IPaymentStrategy / DI registration	TC-NFR-2	Planned
NFR3	E Owner 524H0191	Responsive views / localisation	Views / Localisation	TC-NFR-3	Planned
NFR4	E Owner 524H0191	Identity / authorisation	Identity / Policies	TC-NFR-4	Planned
NFR5	D/F Owner 524H0195 / 524H0179	Append-only history	OrderStatusHistory	TC-NFR-5	Planned
NFR6	D/F Owner 524H0195 / 524H0179	State machine / price snapshot	OrderStateMachine / PricingEngine	TC-NFR-6	Planned